

ApexPay

The all-in-one Financial Operating System for Ethiopia

AI-native payments · Real-time double-entry General Ledger · Workforce & Payroll · Procurement, Tax, FX & Budgeting · Embedded Finance · AI Assistant

Detailed Investor Overview — 2026

Executive Summary

43

Business modules

10/10

CI gates green

29

E2E smoke sections

96.4%

Payment success rate

2.9%

Average take rate

5.2x

LTV : CAC

- **AI assistant built in.** A role-scoped Apex Assistant (RAG + tool-calling agents) answers finance, inventory and payroll questions for merchants, employees, vendors and employers — in English and Amharic.
- **Ethiopia-native compliance.** NBE ONPS/10/2025, ET income-tax brackets, pension 7%/11%, VAT/TOT, Fayda ID, Ethiopian calendar, labour proclamation 1156/2019.
- **Live proof, not mockups.** The merchant dashboard is wired to a real seeded API; a 29-section end-to-end smoke runs green on real Postgres in CI.
- **Production-grade.** Money is always decimal. Decimal; double-entry journals must balance; audit logs are append-only; FIN data is hash + last-4 only.

Market & Problem

Ethiopia's economy is digitizing rapidly, yet businesses still juggle a payment gateway, a payroll tool, spreadsheets, and desktop accounting software that never talk to each other. A single transaction is re-entered across systems, reconciliation is manual, payroll tax is error-prone, and there is no unified view of cash, cost, or compliance.

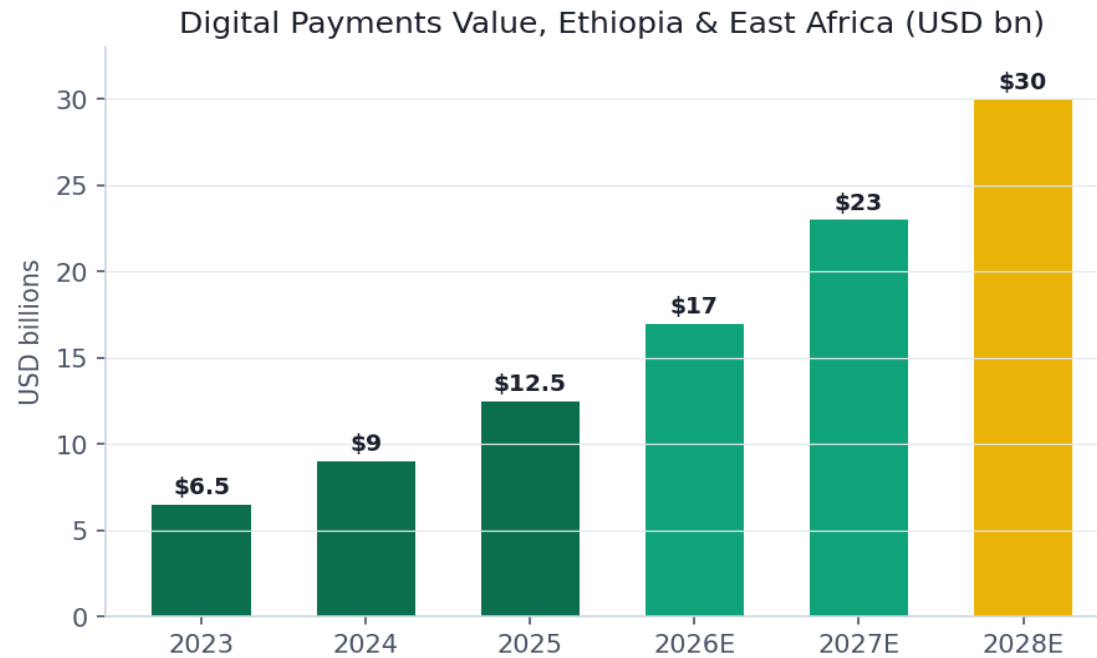
Pain points we solve

- **Fragmentation** — payments, payroll, accounting and procurement live in disconnected silos.
- **Manual reconciliation** — no shared ledger means cash never ties out automatically.
- **Compliance burden** — NBE, tax, pension and labour rules are complex and change often.
- **Inflexible pricing** — global gateways charge FX + cross-border fees on local ETB rails.
- **No data gravity** — no single source of truth for finance teams or AI to reason over.

The macro tailwind is strong: mobile money and digital payments are compounding, and formal MSME formalization is accelerating. ApexPay captures the integrated-finance opportunity rather than any single slice.

Market Size

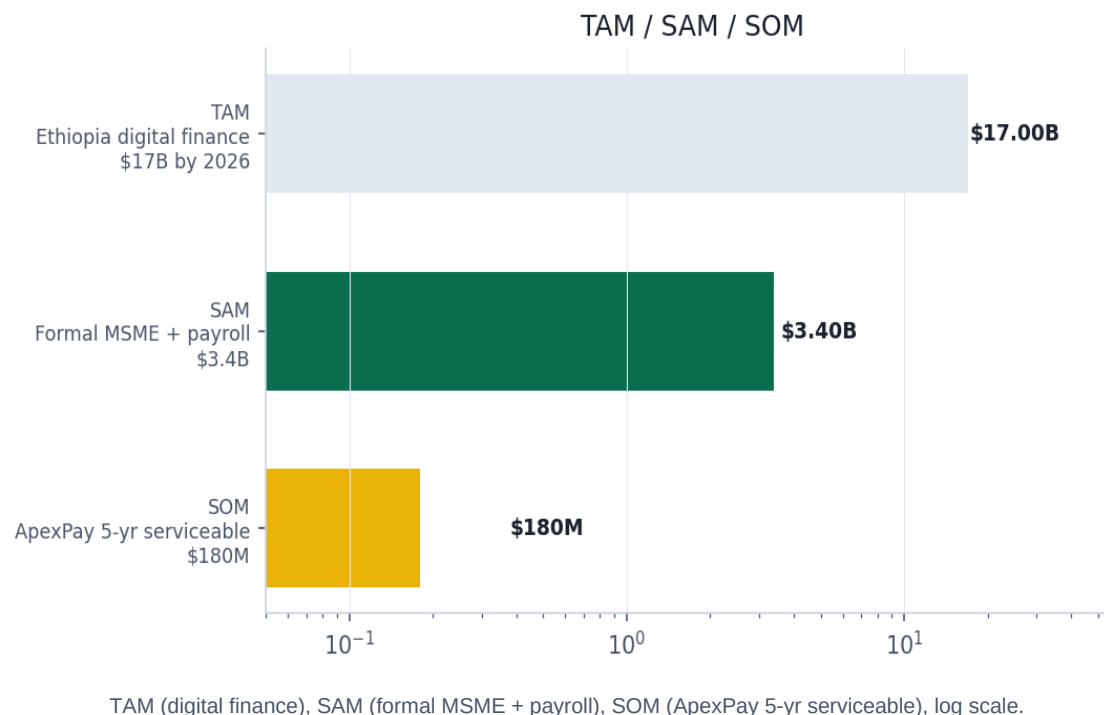
Digital payments across Ethiopia and the wider East African corridor are growing at a high-teens CAGR. ApexPay targets the ETB-denominated domestic rails (Telebirr, CBE Birr, bank IPS) that global gateways under-serve.



Digital payments value, Ethiopia & East Africa, 2023–2028E (USD bn). Sources: NBE, industry reports; illustrative.

Market Sizing — TAM / SAM / SOM

We size a TAM of roughly \$17B for Ethiopian digital finance by 2026. The addressable slice we can realistically serve is formal MSMEs plus payroll (SAM ≈ \$3.4B); our 5-year serviceable market (SOM) is ~\$180M of revenue opportunity, giving ample headroom versus near-term projections.



Solution — One Platform, One Ledger

ApexPay is a multi-tenant platform: one Go API core, one PostgreSQL ledger, four client surfaces (Merchant Web, Admin Web, Checkout Web, Mobile) plus a self-service employee portal. Every module posts into the same double-entry ledger, so numbers always tie out.

Core platform facts

- **API-first** — REST under /v1, typed Go services, pgx + decimal (no float money).
- **Real-time double-entry GL** — journals, fiscal period close, depreciation, inventory COGS, tax schedules, multi-currency FX revaluation.
- **Event-driven** — outbox pattern + workers for reliable async jobs (webhooks, dunning, recon, notifications).
- **Four client surfaces** — Merchant Web, Admin Web, Checkout Web, Mobile (Flutter, Android APK built in CI).
- **Developer portal** — real, DB-backed API-key management and webhook endpoints.
- **i18n** — per-user English/Amharic across API and UI.

Product — Payments

Payments are the wedge. ApexPay initializes a transaction, routes it to the best local rail, enforces 2FA above the NBE threshold, verifies through the connector, and posts the settlement journal to the ledger in the same transaction — so a success is never recorded without its balancing double-entry.

ETB rails

Telebirr · CBE Birr · bank IPS · card · EthSwitch QR

2.9%

Average take rate

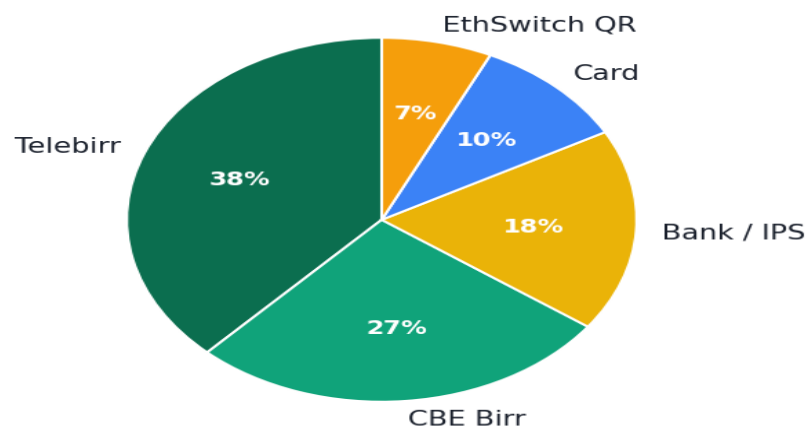
2FA

Enforced above threshold

Idempotent

Safe retries, no double charge

Payment Method Mix (Ethiopia)



Illustrative payment method mix for Ethiopia. ApexPay is rail-agnostic and routes to the healthiest rail per transaction.

Product — Real-time General Ledger

The ledger is the core invariant that differentiates ApexPay from a payments-only gateway. Every business event posts a balanced journal into the merchant's operating book.

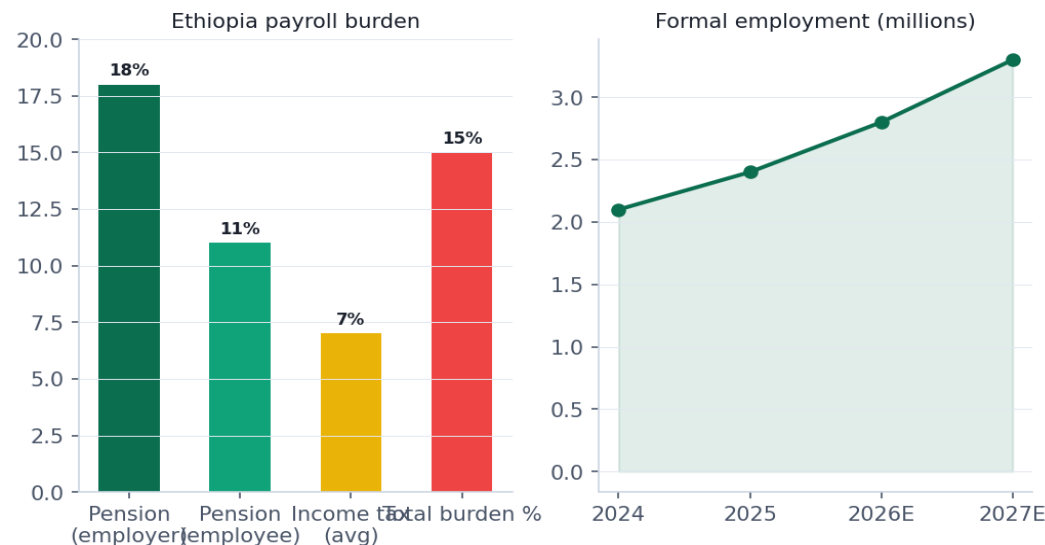
- **Double-entry** — every journal must balance (debits == credits) via `ValidateBalanced` before insert.
- **Idempotent posting** — (`book_id`, `posting_key`) unique so re-posting is safe.
- **Balance integrity** — `ledger_balances` upserted under an advisory lock per book.

- **Append-only** — journal entries are never updated/deleted; audit_logs is now truly append-only (DB trigger).
- **Modules post into one operating book** — payments, refunds, payouts, payroll, depreciation, inventory COGS, tax liability, FX revaluation, expense claims, manual journal entries.
- **Fiscal periods** — close a period to freeze postings; reopen is an explicit operator action.

Product — Workforce & Payroll OS

ApexPay turns payroll into a full workforce operating system: departments, designations, grades, branches, salary structures, employees, payroll runs, attendance, variable inputs, loans, leave, claims, final settlement, and compliance reports.

- **Compliance-native** — ET income-tax brackets, pension employee 7% / employer 11%, labour proclamation 1156/2019 (leave, severance, final settlement).
- **Formula engine** — secure tokenization + shunting-yard; OT 1.25x/1.5x/2.0x per Art 90.
- **Run lifecycle** — create → attendance → calculate → approve (maker-checker) → disburse → ledger + bank/pension/ERCA files.
- **Employee portal** — own payslips, leave, claims; magic-link JWT 24h.



Ethiopia payroll burden (left) and formal-employment growth (right) — the compliance complexity ApexPay automates.

Product — Procurement, Inventory, Tax & FX

Beyond payments and payroll, ApexPay covers the full working-capital loop so finance never leaves the platform.

- **Procurement / AP** — vendors, purchase orders (line items + tax), goods receipts, AP invoices with PO matching, and AP aging buckets.
- **Inventory / Sales** — products, stock movements, orders, COGS posts to the ledger.
- **Tax** — VAT/TOT/withholding schedules, tax register, automatic ledger posting, automated challans.
- **FX** — multi-currency accounts, FX revaluation to ETB, gain/loss posted to GL.
- **Budget / FP&A** — budgets vs actual variance per category.
- **Invoicing** — issue/send invoices, aging, auto-post collected VAT into the tax register.

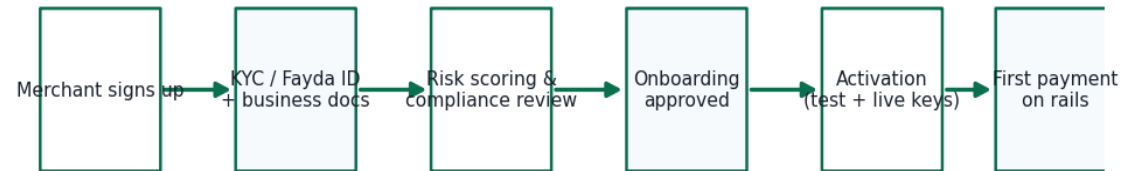
Product — Embedded Finance

Embedded finance turns operational data into lending and banking products, all reconciled to the same ledger.

- **Credit lines & loans** — collateral-free working-capital credit lines, loan disbursements, repayment scheduling, credit scoring from TPV/payroll data.
- **Escrow** — automated marketplace P2P holds and release; platform fee + withholding tax.
- **Corporate cards** — virtual/physical, dynamic limits up to 2Cr ETB equivalent, spending controls, 1% cashback, 2.5% forex markup.
- **Current & virtual accounts** — smart collections via virtual accounts, multi-currency current accounts.
- **Forex & credit** — FX rates/requests, credit lines with utilization tracking.

User Flow — Merchant Onboarding

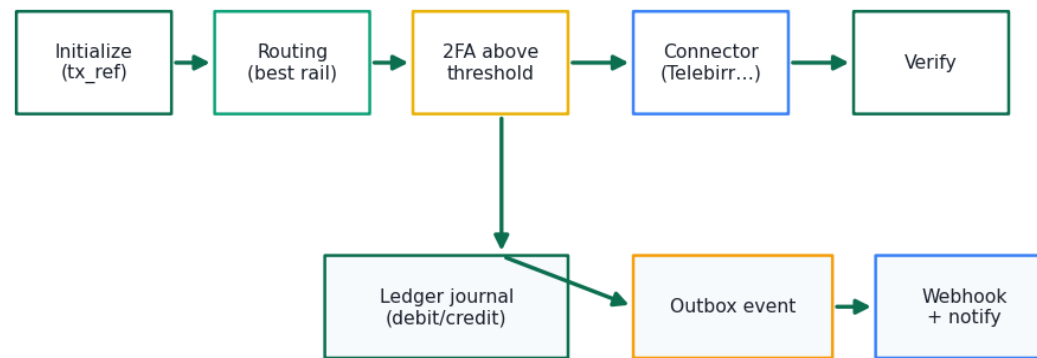
Merchant Onboarding Flow



End-to-end onboarding: signup → KYC/Fayda → risk → approval → activation → first payment.

User Flow — Payment Lifecycle

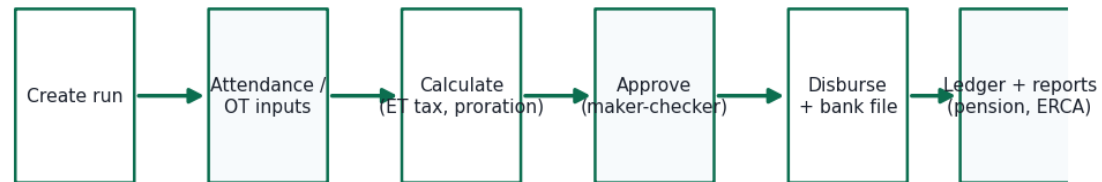
Payment Lifecycle (one transaction, one ledger)



One transaction: initialize → route → 2FA → connector → verify → ledger journal → outbox → webhook.

User Flow — Payroll Run

Payroll Run Flow (compliance: ET tax + pension 7%/11%)



Payroll run: create → attendance/OT → calculate → approve → disburse → ledger + compliance reports.

App — Payments (live data)

Real screenshot of the running payments page showing transactions with amount, method, connector, 2FA, and status badges.

ApexPay

Dashboard

Payments

Refunds

Payouts

Payroll

Workforce

Subscriptions

Banking

Treasury

Invoices

Risk & Fraud

Analytics

Accounting

Inventory/Sales

Disputes

Loyalty

Lending

Fixed Assets

Compliance

production-et

EN

Sign out

Payments

Tx Ref	Amount	Method	Status	Connector	2FA	Action
txr_refund_smoke	ETB 250	mock	succeeded	mock	-	View •

App — Subscriptions (live data)

Real screenshot of the running subscriptions page showing customer, plan, amount, status, and period end from the seeded DB.

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Sign out

Subscriptions

Subscriptions

Customer	Plan	Amount	Status	Period End	Created
cust_smoke	splan_smoke	—	active	2026-08-21T18:16:44.021197Z	0001-01-01

App — Payouts (live data)

Real screenshot of the running payouts page showing a seeded payout batch (PBATCH-SMOKE-001) and its status.

ApexPay

production-et

EN

Sign out

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Compliance

Single Payout • 0000 0000 — Maker-checker >50k ETB

Abebe Kebede • CBE ****1234 veriv

10000

pout_ref_001

Create Payout • pending_approval if >50k • Ledger M3 Dr merchant_payable Cr clearing bank atomic per batch book

Balance check insufficient: balance 400 • ApprovalThreshold 50k • GetMerchantBalance COALESCE sum net_amount succeeded - sum payouts queued/processing/succeeded

Bulk CSV Upload • 0000 — 1000 rows preview outstanding • GitHub Actions timeline • Papaparse Real Streaming O(n)

Drop CSV here • 0000 000000 • name,account_no,bank_code,amount,payout_ref
Preview table validation icons green/red • Amount sum fees calc MDR 2.9% • Timeline like GitHub Actions steps • Levenshtein fuzzy name match <3

Choose File No file chosen

Real: Papaparse streaming O(n) no OOM per Day 3 + Levenshtein DP O(n*m) optimal per banking verification name_match + balance check sufficient + maker-checker required

Name	Account	Bank	Amount	Ref	Status
Abebe	****6789	CBE • Commercial Bank of Ethiopia	ETB 10000	pout_ref_01	valid
Almaz	****6790	AWASH • Awash Bank	ETB 5000	pout_ref_02	valid
Kebede	****6791	DASHEN • Dashen Bank	ETB 15000	pout_ref_03	warning name mismatch Levenshtein 2 require override note

Total ETB 30,000 • 3 rows • 2 valid • 1 invalid/warning • Maker-checker required • Balance check sufficient • Fee calc MDR 2.9%

Create Batch • pbat_01H • pending_approval • Dual approve >50k • Ledger M3 Dr payable Cr clearing total per batch book + CreateBulkTx atomic batch+payouts+journal+balances

App — Banking / Current Accounts (live data)

Real screenshot of the running banking page showing current accounts (ETB-CBE-7778889990) and balances from the seeded DB.

ApexPay — Confidential Investor Deck

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ApexPay

Dashboard

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Payouts

Payroll

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production-et

EN

Sign out

Current Accounts

Real partner-bank current accounts (CBE/Awash/Dashen) with balances and card/cheque status.

Account #	Name	Type	Bank	Status	Balance	Primary
USD-CBE-1000001	Docker Smoke USD	current	Commercial Bank of Ethiopia	active	ETB 10000.00000000	—
ETB-CBE-7778889990	Docker Smoke PLC	current	Commercial Bank of Ethiopia	active	ETB 125000.00000000	—

App — Lending (live data)

Real screenshot of the running embedded-finance lending page showing a seeded loan (loan_smoke).

ApexPay

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Disputes

Loyalty

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EN

Sign out

Request Loan

Credit line ID

Amount ETB

Working capital

Request Loan

ID	Amount	Purpose	Rate	Outstanding	Status
loan_snoke	ETB 500000.00000000	inventory	18.00%	ETB 500000.00000000	disbursed

App — Developer Portal (live data)

Real screenshot of the running developer portal showing API keys with scopes and status.

ApexPay — Confidential Investor Deck

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ApexPay

Dashboard

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Invoices

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Compliance

production-et

EN

Sign out

API Keys

Test/Live keys, scoped, hash-at-rest. Secrets shown once.

Refresh

Create a key

Name

e.g. mobile app

Environment

Test

+ Create Key

Your keys

integration • sk_test_demo***** • test • secret

active

Revoke

Scopes: role:ops • Last used 8/11/2026, 6:24:31 PM

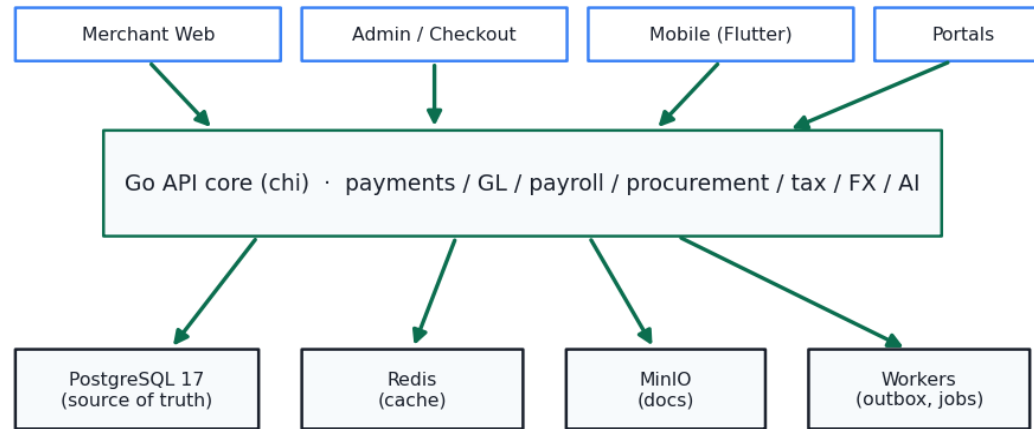
Public Key & Embedded SDK

Use the public key with checkout.js — safe for the frontend, no secret.

```
<script src="https://checkout.apexpay.et/sdk.js"></script>
<script>
const apexpay = new ApexPay('pk_test_...');
apexpay.checkout({ amount: '500', currency: 'ETB', tx_ref: 'txr_' + Date.now(),
method: 'telebirr', customer_email: 'cust@example.et',
return_url: 'https://example.et/return', callback_url: 'https://example.et/callback' });
</script>
```

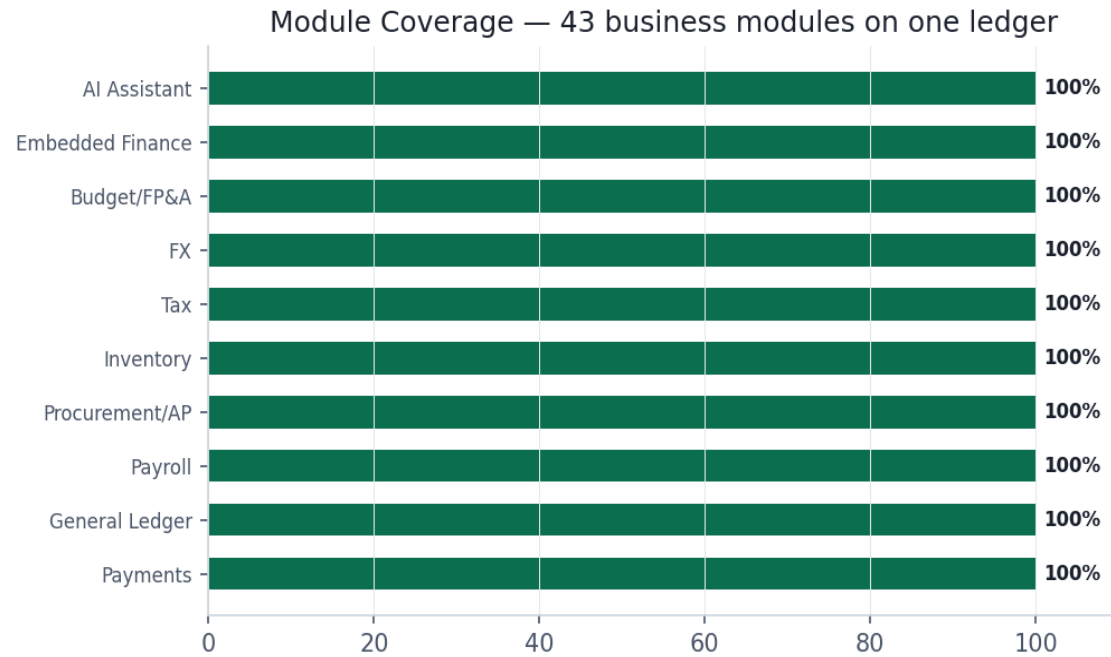
System Architecture

System Architecture — one money core, thin clients



One Go API core, one PostgreSQL ledger, Redis cache, MinIO vault, and background workers — with thin client surfaces.

Module Coverage



All 10 core capability groups are implemented and share one ledger — 43 modules total.

AI Assistant — Role-Scoped, Grounded, Bilingual

The Apex Assistant authenticates the caller, resolves whether they are a merchant, employee, vendor or admin, and answers only within that scope — grounded in live ledger/data via tools, never hallucinated.

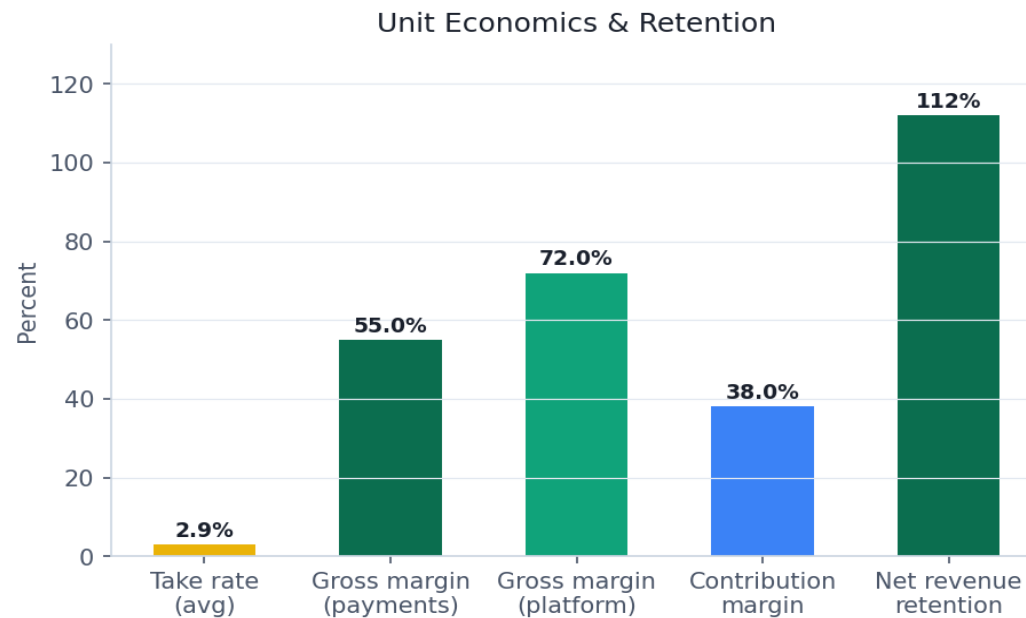
- **Tool-calling agents** — deterministic intent routing (ADR-006) to read-only, actor-gated tools against live data.
- **RAG with mandatory citations** — grounded answers with a confidence threshold; below threshold, no answer.
- **Bilingual** — English + Amharic via the i18n catalog.
- **Append-only history** — full conversational audit trail.
- **Examples** — 'What is my cash position and TPV today?', 'Show the balance sheet', 'My leave balance'.

Security, Quality & Compliance

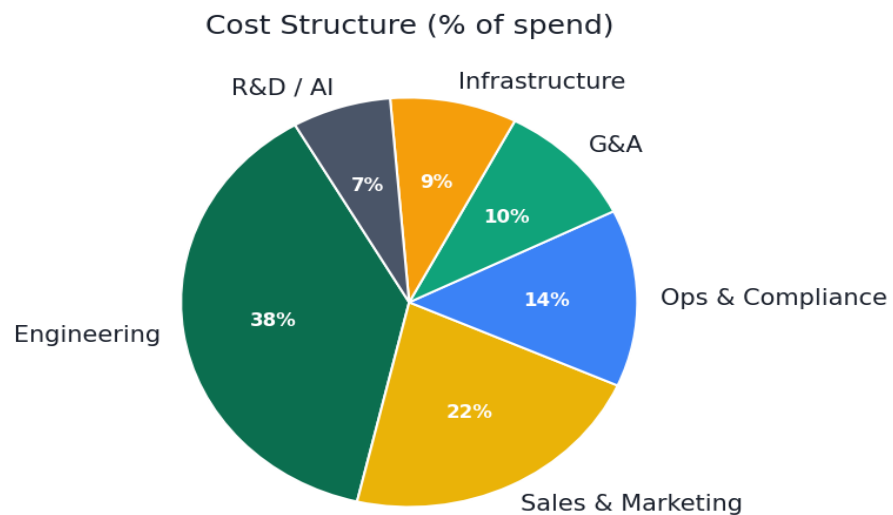
ApexPay is engineered for money safety and auditability, and enforces it in CI.

- **Money** — always decimal.Decimal; no float; double-entry validated on every journal.
- **Secrets** — API keys hashed at rest (sha256), prefix+digest verification; connector keys AES-GCM; webhook secrets encrypted; gitleaks + gosec + trivy in CI.
- **FIN privacy** — FIN stored as salted hash, only last-4 returned/logged.
- **Network** — webhook endpoints are SSRF-protected (https-only, private IP blocked).
- **Auditability** — append-only audit_logs (DB trigger) and append-only assistant history.
- **Compliance** — NBE ONPS/10/2025, ET income-tax, pension 7%/11%, VAT/TOT, Fayda, labour 1156/2019.

Unit Economics & Cost Structure



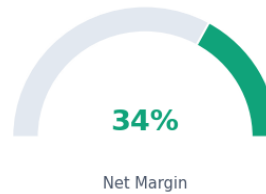
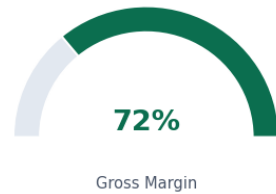
Take rate, gross margin, contribution margin, and net revenue retention (112%) — platform economics improve with scale as embedded finance layers on.



Indicative cost structure — engineering-led, with a funded go-to-market to capture the land-and-expand motion.

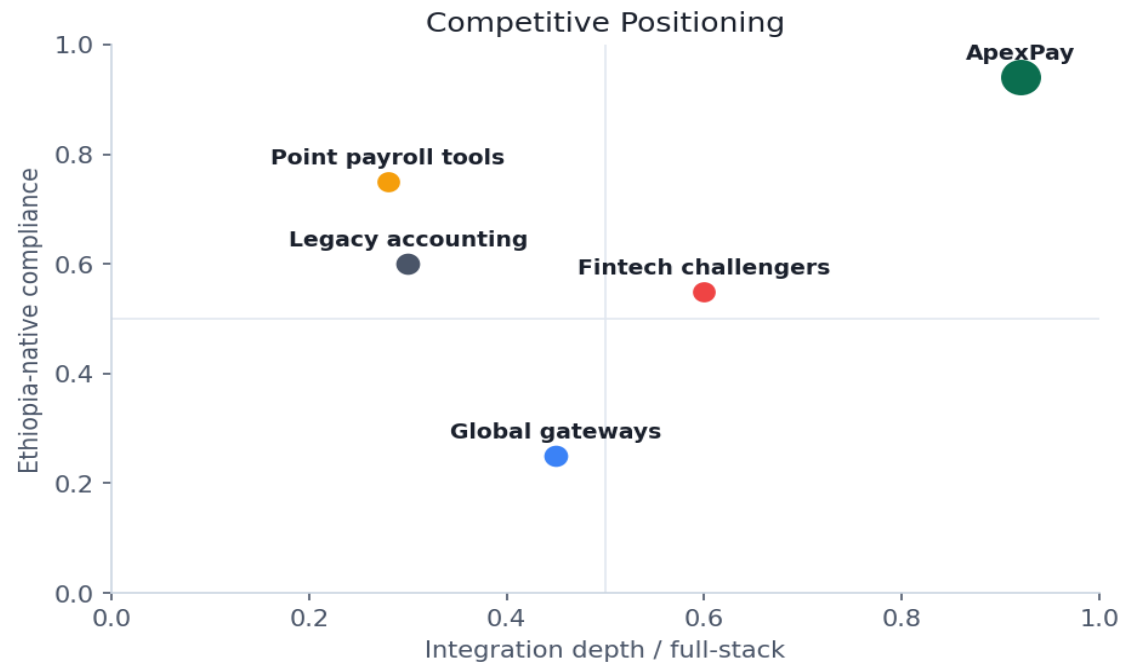
Key Ratios & KPIs

Key Ratios & KPIs



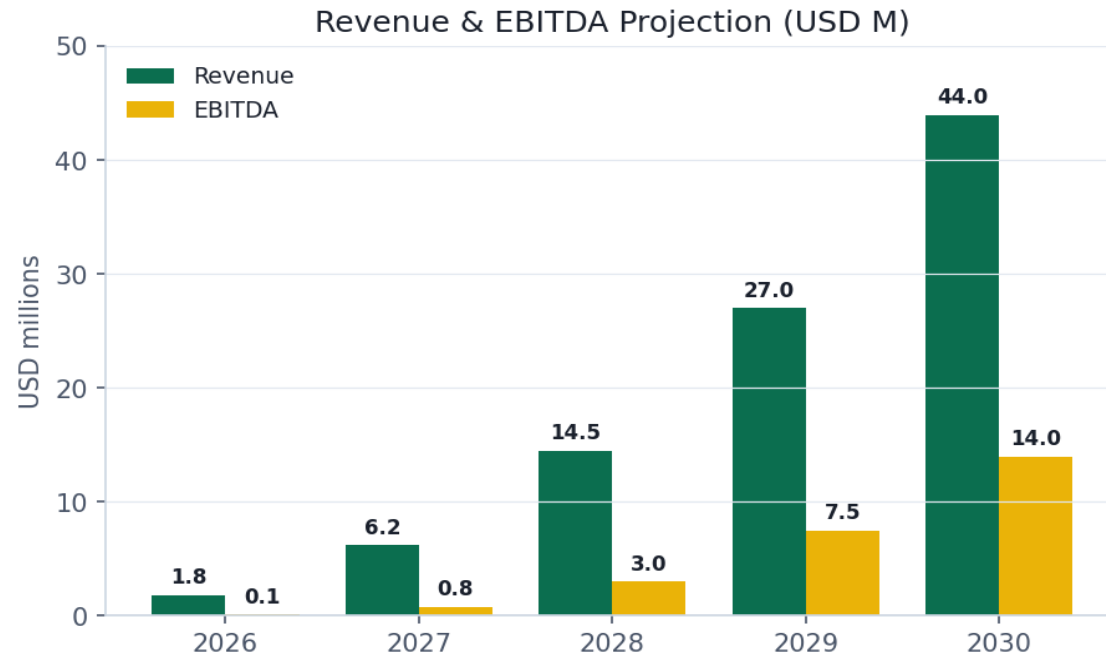
Gauges for margin, CAC payback, LTV:CAC, burn, and runway.

Competitive Positioning



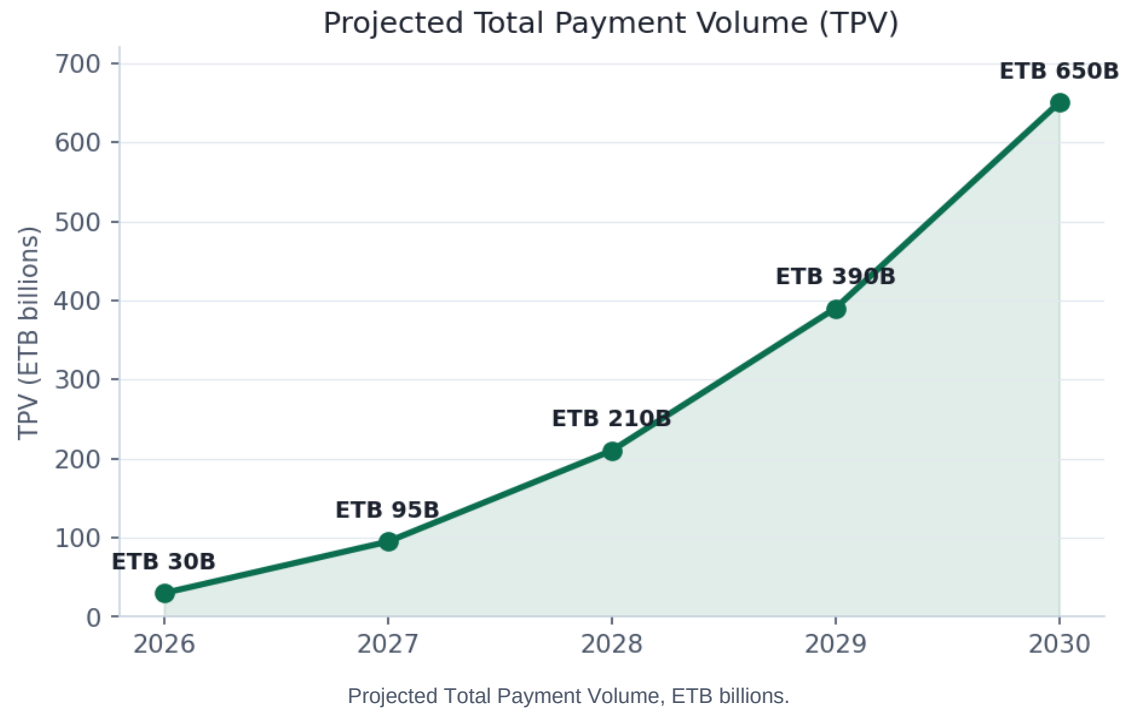
ApexPay combines full-stack integration with Ethiopia-native compliance that point solutions and global gateways don't offer.

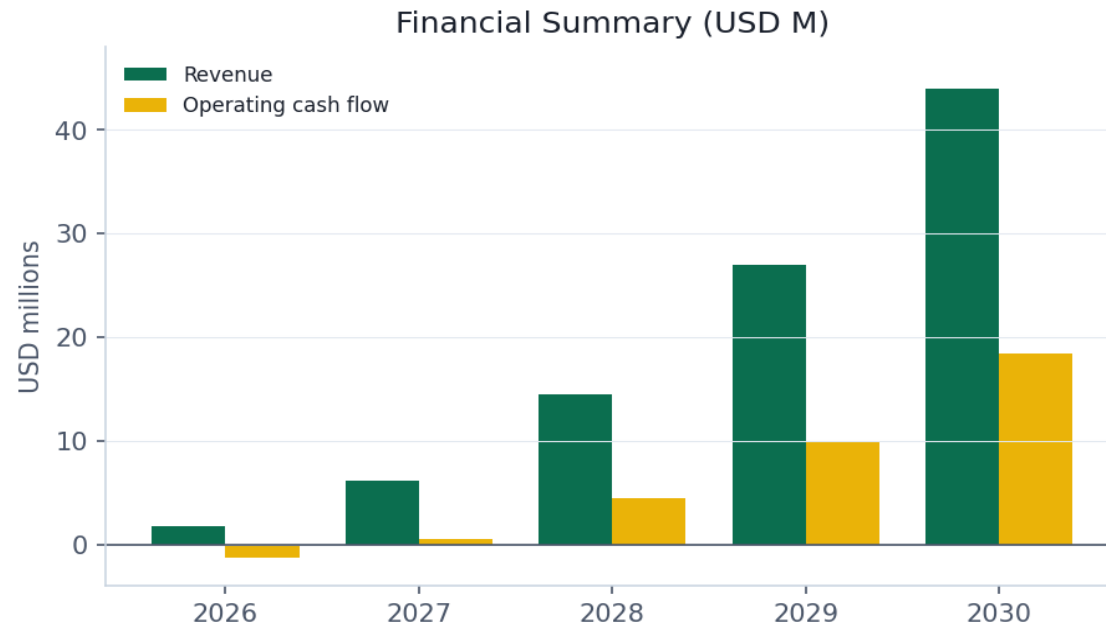
Financial Projections



Revenue and EBITDA ramp (USD M), 2026–2030.

TPV Growth & Financial Summary





Revenue vs operating cash flow (USD M).

Delivery Roadmap

Delivery Roadmap



From core payments to embedded finance and scale — every milestone shipped working software with CI green.

Team & The Ask

ApexPay is built and operating as a live, verified product. We are raising to accelerate go-to-market and deepen the embedded-finance and AI moat.

Live

Product running on real data

43

Modules shipped

4

Client surfaces

1

Unified ledger

Use of funds

- **Go-to-market** — merchant acquisition across MSME segments (30%).
- **Embedded finance** — credit underwriting, banking partnerships (25%).
- **Engineering & AI** — assistant depth, scale, reliability (30%).

- **Compliance & ops** — NBE licensing, audits, security (15%).

Risk & Mitigations

Risk	Mitigation
Regulatory change	Deep NBE/ONPS compliance, licensing track, adaptable rails
Credit risk (embedded finance)	Credit scoring from TPV/payroll data, exposure caps, ledger-settled
Competition	Integrated ledger + AI + local rails is hard to copy feature-by-feature
Scope sprawl	Phased roadmap, finish-before-add guardrail, single shared ledger
Multi-tenancy isolation	Every query scoped by merchant_id; portals by entity
Operational	Docker/distroless, health checks, worker idempotency, reconciliation

